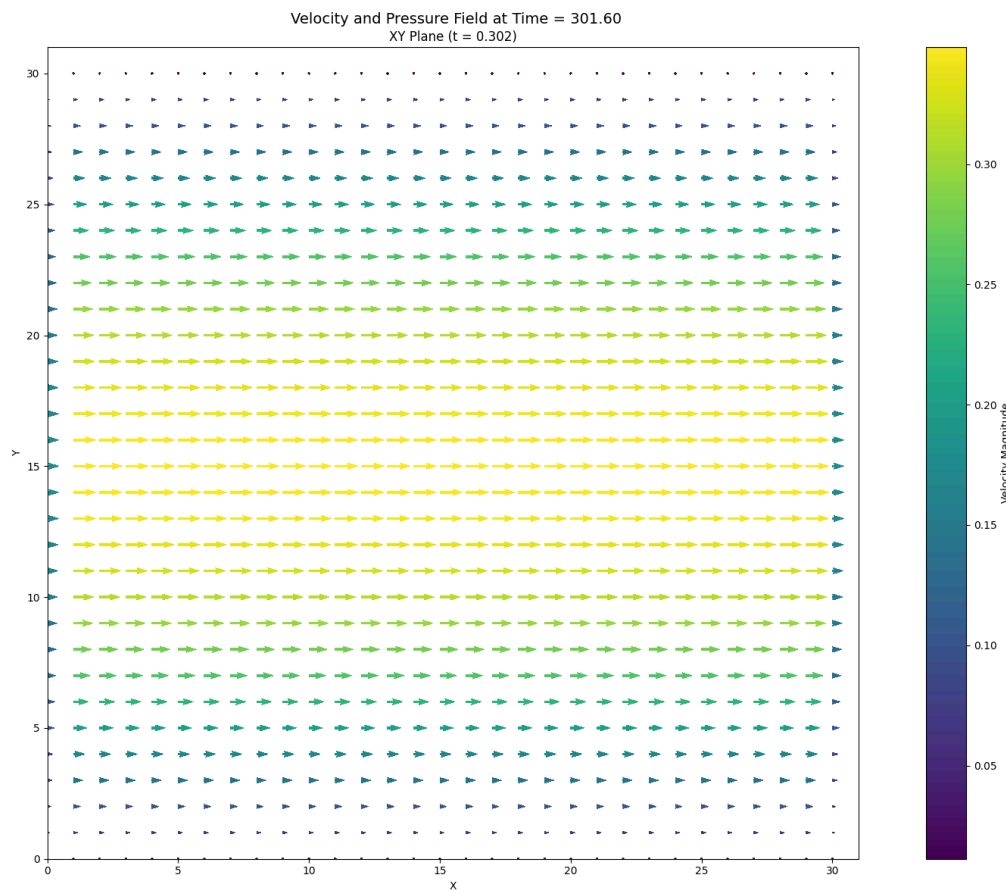


Experimental Report

Case: Poiseuille Flow

Description of Flow: This case models a classical plane Poiseuille flow in a channel driven by a constant streamwise body force. The fluid is confined between two parallel no-slip walls, and the imposed forcing balances viscous diffusion to produce a smooth parabolic velocity profile along the wall-normal direction. The goal is to verify that the learned PDE reproduces this body-force-driven laminar flow from the saved field data.

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.



1. Governing Equation Verification

Reference equation

$$u_t = 8.80434783e-05 u_{yy} + 9.81000000e+00 \text{one}$$

Learned equation

$$u_t = 8.84726274e-05 u_{yy} + 9.85781675e+00 \text{one}$$

Term	Reference coefficient	Learned coefficient	Absolute error
$u*u_y$	$+0.00000000e+00$	$+0.00000000e+00$	$0.00000000e+00$
$u*u_{yy}$	$+0.00000000e+00$	$+0.00000000e+00$	$0.00000000e+00$
u_y*u_{yy}	$+0.00000000e+00$	$+0.00000000e+00$	$0.00000000e+00$
u	$+0.00000000e+00$	$+0.00000000e+00$	$0.00000000e+00$
u_y	$+0.00000000e+00$	$+0.00000000e+00$	$0.00000000e+00$
u_{yy}	$+8.80434783e-05$	$+8.84726274e-05$	$4.29149147e-07$
1	$+9.81000000e+00$	$+9.85781675e+00$	$4.78167517e-02$

2. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	ρ	$9.20000000e+02$	kg m^{-3}
Dynamic viscosity	ν	$8.10000000e-02$	Pa s
Kinematic viscosity	ν	$8.80434783e-05$	$\text{m}^2 \text{s}^{-1}$
Characteristic velocity	u_{max}	$1.00000000e+00$	m s^{-1}
Characteristic length	D	$5.00000000e-03$	m
Reynolds number	Re	$1.97740055e+01$	dimensionless
Body force	G	$9.81000000e+00$	m s^{-2}
Grid points	N_x	$3.10000000e+01$	points
Unused transverse grid points	N_y	$3.10000000e+01$	points
Grid spacing	Δy	$1.61290323e-04$	m
Time step	Δt	$1.31646880e-04$	s
Numerical threshold	ϵ	$1.00000000e-15$	-

3. Numerical Discretisation

Quantity	Symbol	Value	Unit
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Parameters listed above are the values used by the selected solver.

4. Data and Pre-processing

Quantity	Value
Loaded field shape (t, y)	$(2143, 31)$
Regression field shape (t, y)	$(543, 27)$

Quantity	Value
Regression samples	14,661
Dictionary terms	7
First saved time index	150
Temporal crop at each boundary	800 layers
x-boundary crop	0 points
y-boundary crop	2 points

5. STRidge Selection and Error Metrics

Metric	Value
Regularisation parameter, λ	1.00000000e-09
Selected tolerance	1.00000000e-06
L0 penalty	1.00000000e-06
Training RMSE	1.11757497e-13
Validation RMSE	1.24716034e-13